10 **(i)** A curve has equation $y = 16x + \frac{1}{x^2}$. Find

(*A*) $\frac{dy}{dx}$, **[2]**

(*B*) $\frac{d^2y}{dx^2}$. **[2]**

(ii) Hence

- find the coordinates of the stationary point,
- determine the nature of the stationary point. **[5]**